

Adaptive DRL-Based Virtual Machine Consolidation in Energy-Efficient Cloud Data Center

Jing Zeng^{ID}, Ding Ding^{ID}, Kaixuan Kang^{ID}, HuaMao Xie^{ID}, and Qian Yin

Abstract—The dramatic increasing of data and demands for computing capabilities may result in excessive use of resources in cloud data centers, which not only causes the raising of energy consumption, but also leads to the violation of Service Level Agreement (SLA). Dynamic consolidation of virtual machines (VMs) is proven to be an efficient way to tackle this issue. In this paper, we present an Adaptive Deep Reinforcement Learning (DRL)-based Virtual Machine Consolidation (ADVMC) framework for energy-efficient cloud data centers. ADVMC has two phases. In the first phase, Influence Coefficient is introduced to measure the impact of a VM on producing host overload, and a dynamic Influence Coefficient-based VM selection algorithm (ICVMS) is proposed to preferentially choose those VMs with the greatest impact for migration in order to remove the excessive workloads of the overloaded host quickly and accurately. In the second phase, a Prediction Aware DRL-based VM placement method (PADRL) is further proposed to automatically find suitable hosts for VMs to be migrated, in which a state prediction network is designed based on LSTM to provide DRL-based model more reasonable environment states so as to accelerate the convergence of DRL. Simulation experiments on the real-world workload provided by Google Cluster Trace have shown that our ADVMC approach can largely cut down system energy consumption and reduce SLA violation of users as compared to many other VM consolidation policies.

Index Terms—Cloud computing, vm consolidation, energy efficient, influence coefficient, deep reinforcement learning

1 INTRODUCTION

As a new technology based on distributed computing, Aggrid computing and virtualization, cloud computing promises to provide users with computational resources (e.g., CPU, RAM, Bandwidth) as well as convenient services such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS) [1]. Along with the rapid development of big data and 5G [2], [3], cloud computing has dramatically changed the IT operations in the industry in recent years.

With the rapid development of cloud computing, the scale of cloud computing facilities expands largely, accompanied by a growing concern about energy consumption [4]. It is reported that cloud data centers alone currently utilize about 7% of all electricity consumption in the world, and this figure is predicted to increase to 13% by the year

2030 [5]. According to the forecast of Japan's Ministry of Economy, Trade and Industry (METI), global IT energy consumption will quadruple by 2025 and increase 12 times by 2050 [6]. The prominent problem of high energy consumption in cloud data centers, on the one hand, increases the operation costs of service providers; on the other hand, it also reduces energy efficiency of the whole society, resulting in a great waste of resources.

Resource utilization in cloud data centers directly affects energy consumption of the whole system [7], [8]. Unreliable characterization of overloaded/underloaded physical hosts will lead to energy growing, performance degradation and SLA violation. For cloud providers, a profitable strategy is to dynamically integrate VMs onto fewer physical hosts to save hardware, and to reduce energy consumption. Among various ways towards energy-efficient cloud data centers, VM consolidation is proven to be one of the most productive techniques to optimize resource utilization and to cut down energy consumption. Most of recent works summarize VM consolidation problem in three sub-processes [9], [10]: 1) workload detection: detecting overloaded/underloaded hosts; 2) VM selection: selecting proper VMs for migration; 3) VM placement: finding suitable hosts for the selected VMs to place. With these steps, VM consolidation has a major impact on energy efficiency through live migration by packing VMs onto the minimum number of hosts and switching the idle hosts to a power-saving mode. Therefore, VM consolidation is the very place where we can explore innovative methods to manage resources and guarantee Quality of Service (QoS) in an energy-aware way.

However, the dynamic cloud environment with constant change of workload poses many challenges on effective VM

- Jing Zeng, Kaixuan Kang, HuaMao Xie, and Qian Yin are with the School of Computer and Information Technology in Beijing Jiaotong University, Beijing 100044, China. E-mail: {19120435, 19112015, 20112033, 19120428}@bjtu.edu.cn.
- Ding Ding is with the School of Computer and Information Technology, Beijing Jiaotong University, Beijing 100044, China, and also with the Beijing Key Lab of Traffic Data Analysis and Mining, Beijing 100044, China. E-mail: dding@bjtu.edu.cn.

Manuscript received 31 Aug. 2021; revised 20 Dec. 2021; accepted 23 Jan. 2022. Date of publication 1 Feb. 2022; date of current version 23 May 2022.

This work was supported in part by the R&D Program of Beijing Municipal Education Commission under Grant KJZD20191000402, in part by the Beijing Natural Science Foundation under Grant L211015, and in part by the Fundamental Research Funds for the Central Universities under Grants 2019JBM025 and 2019JBZ104.

(Corresponding author: Ding Ding.)

Recommended for acceptance by A. J. Peña, M. Si and J. Zhai.

Digital Object Identifier no. 10.1109/TPDS.2022.3147851

consolidation. Most existing methods suffer from poor flexibility in strategy adjustment and are difficult to achieve long-term optimal management goals. Therefore, the objective of this study is to investigate a dynamic VM consolidation mechanism based on deep reinforcement learning (DRL) technique where VM can be selected and placed appropriately and adaptively pertinent to the workload status of hosts. The main contribution of this work can be summarized as follows:

- 1) We have proposed an Adaptive DRL-based Virtual Machine Consolidation (ADVMC) framework for energy-efficient cloud data centers where VM selection and VM placement are both implemented to reduce energy consumption and to drop SLA violation.
- 2) A Influence Coefficient (IC) is introduced to measure the impact of a VM on producing host overload, and a dynamic Influence Coefficient-based VM selection algorithm (ICVMS) is also explored to preferentially choose those VMs with the greatest impact for migration, which helps to get rid of excessive workloads quickly and accurately for overloaded hosts.
- 3) A Prediction Aware DRL-based VM placement method (PADRL) is developed to find suitable hosts for VMs to be migrated in an energy efficient way. DRL-based model enables an adaptive means to obtain optimal policy automatically. Besides, a state prediction network is designed based on LSTM to provide the model more reasonable environment states to accelerate the convergence of DRL.
- 4) Extensive experiments on a real-world workload have confirmed the effectiveness of the proposed method, and the results demonstrate that our ADVMC, credited to both VM selection and VM placement, outperforms many state-of-the-art models significantly.

The remainder of this paper is organized as follows. Related works are discussed in Section 2. The overall framework of energy aware VM consolidation is given in Section 3. The Influence Coefficient-based VM selection and the Prediction Aware DRL-based VM placement are introduced in Sections 4 and 5 respectively. In Section 6, we present the simulation experiments and performance evaluation. In Section 7 we conclude the paper.

2 RELATED WORKS

In cloud data centers, an appropriate VM consolidation has become an effective method to improve resource utilization, and to reduce energy consumption and operational costs of systems. Thus, VM consolidation has received much attention and has been studied extensively in recent years with the key objective to tightly package VMs while preserving the performance and the SLA.

Many existing researches have formulated the VM consolidation as a well-known bin packing problem [11], [12]. This problem is known to be NP-hard, and several heuristic algorithms have been utilized to approximate the optimal solution, such as First Fit (FF), First Fit Decreasing (FFD), Best Fit (BF) and Best Fit Decreasing (BFD) [13], [14]. Besides, Beloglazov *et al.*, [15] developed a Power Aware Best Fit

Decreasing (PABFD) algorithm to place VM on a host which will increase least power consumption after allocation. Khoshkholghi *et al.* [16] proposed a VM consolidation algorithm to trade off energy and performance, in which all VMs from underloaded hosts would be migrated and the idle hosts would be switched to an energy-saving mode with fast transition times to reduce total energy consumption. However, most of heuristic algorithms can not guarantee effective allocation and do not work well for large scale cloud data centers.

A lot of methods have been proposed to solve the VM consolidation problem as an optimization problem using meta heuristic algorithms. In [17], Guo *et al.* proposed an energy-efficient fault-tolerant static scheduling algorithm (EFTP) to realize the fault tolerance, and to reduce energy consumption based on extended particle swarm optimization (PSO) approach. In [18], Kansal *et al.* proposed a firefly algorithm-based method to maintain the performance and energy efficiency of the cloud data centers by migrating the maximally loaded virtual machine to the least loaded active node. In [19], Zhang *et al.* proposed a multi-objective optimization model of fault-tolerant initial VM placement based on a heuristic ant colony algorithm to solve the reliability problem in cloud computing systems. However, meta heuristic algorithms not only require pre-defined knowledge based on which allocation is done, but also suffer from high variance and poor convergence.

As the rapid development of machine learning and deep learning, many workload prediction approaches have been studied for resource management and power management in large-scale cloud systems. Farahnakian *et al.* [20] presented a Utilization Prediction-aware VM Consolidation (UP-VMC) approach in which a regression-based model is used to approximate the future CPU and memory utilization of resources in order to reduce energy consumption, the number of VM migrations and the number of SLA violations. Hieu *et al.* [21] presented a virtual machine consolidation algorithm with multiple usage prediction (VMCUP-M) to estimate the long-term utilization of multiple resource types based on the local history of the considered servers, thereby reducing the number of migrations and the power consumption after consolidation. Some well-known time series prediction method have also been introduced for the dynamic consolidation of VMs in cloud data centers. For example, Gupta *et al.* [22] proposed to use multivariate Long Short Term Memory (LSTM) models for prediction of resource usage to achieve optimal resource provisioning in cloud. These workload prediction-based VM consolidation methods rely largely on the accuracy of the forecast.

However, most of these methods mentioned above may not adaptively adjust strategies well in response to the changes of application scenarios and may result in redundant consolidation decisions. Reinforcement Learning (RL) is a kind of machine learning technique which enables an intelligent agent to learn the optimal policy by interacting with the environment [23]. Compared with traditional machine learning methods, RL can make decisions efficiently for its excellent self-learning ability and thus is more suitable for the area of VM consolidation [24]. Recent researches employing RL approach have demonstrated the effectiveness and benefits of adaptive learning for autonomic resource management in cloud [9]. As a kind of intelligent system from environmental

state to behavior, RL enables agent to learn in a highly stochastic, non deterministic environment with no prior knowledge of its surrounding environment [25]. In this context, the VM consolidation problem is regarded as a Markov sequence decision problem in which every decision action is only related to the current system state, and has nothing to do with the past. D. Basu *et al.* [26] proposed an online RL algorithm, named Megh, to learn uncertain dynamics. It can work in real-time without incurring significant execution overhead. Our previous study in [27] further proposed a two-phase energy efficient framework by using M/M/S queuing model and Q-learning to distribute resource requests, intending to improve server response time as well as CPU utilization and energy consumption in cloud computing systems. Although traditional reinforcement learning approach can optimize metrics of energy consumption and SLA in resource management system, the tabular value-based RL methods still face the problem of limited scalability.

To address the problem of huge state-action space in large scale cloud data center, deep reinforcement learning-based methods like Deep Q-Learning (DQN) have proposed for efficient resource management in recent years. Liu *et al.* [28] presented a novel hierarchical framework to solve the problem of high dimensions in state and action space for optimizing resource utilization and improving energy efficiency. Zhang *et al.* [29] presented an energy-efficient scheduling scheme based on deep Q-learning model (DQL-EES) for periodic tasks in real-time systems. DQL-EES was designed by combining a stacked auto-encoder and a Q-learning model in which the stacked auto-encoder is used to replace the Q-function for learning the Q-value. Cheng *et al.* [30] developed a DRL-based two-stage RP-TS processor to automatically generate the best long-term decisions by learning from the changing environment, so as to minimize energy cost. However, aggressive VM consolidation may degrade the performance. Therefore, a performance energy trade-off between providing high-quality service and reducing system energy consumption is highly desired.

3 ENERGY AWARE VM CONSOLIDATION FRAMEWORK

In this part, the system model is first illustrated in Section 3.1, and the overall framework of our energy aware VM consolidation is further defined in Section 3.2.

3.1 System Model

In IaaS environment, cloud providers provide users with virtualized computing resources through the Internet. Thus, the cloud system presented in this paper is composed of a large number of physical hosts with virtual machines. We frame the context as follows. The hosts are indexed by i where $1 \leq i \leq N$. Let H be the set of physical hosts and thus H can be represented by $H = \{H_1, H_2, \dots, H_N\}$. Each host H_i contains a fixed number of n_i of virtual machines, represented by $V_{i,1}, V_{i,2}, \dots, V_{i,n_i}$.

3.1.1 Energy Consumption

In general, power consumption of a physical host in cloud data centers is closely related to its CPU utilization [15]. In

this paper, the real data on power consumption provided by the results of the SPECpower benchmark [31] is adapted instead of using an analytical model of server power consumption. Hence, the energy consumption can be measured based on provided power consumption. Supposing that $P_i(t)$ is the power consumption of host H_i at time t , then the energy consumption of host H_i for a time period of T can be given by:

$$E_i(T) = \int_0^T P_i(t)dt, \quad (1)$$

and the energy consumption of the whole data center related to time t can be formulated as the sum of energy consumption by all hosts:

$$E_t = \sum_{i=1}^N E_i(T), \quad (2)$$

where N is the number of hosts in cloud data center.

3.1.2 Migration Cost

Live migration of VMs allows transferring a VM between physical nodes without suspension and with a short downtime. However, live migration has a negative impact on the performance of applications running in a VM during a migration. Research has shown that the length of a live migration usually depends on both the total amount of memory used by the VM and available network bandwidth [32]. But for the class of applications with variable workloads, the average performance degradation including the downtime can be estimated as approximately 10% of the CPU utilization [15]. Therefore, 10% of the CPU utilization is introduced as default measure of migration cost in CloudSim and is often adopted by many VM consolidation studies experimented on CloudSim [20], [21]. In this paper, it is also used for migration cost as many others, which has been further involved in our CPU utilization-related energy consumption by increasing 10% CPU utilization for VM each time live migration occurs.

3.1.3 SLAV

SLAV (Service Level Agreement Violation) is often used to measure the ability of cloud providers to deliver service guarantees. Here, it is used as another important indicator to measure user satisfaction of service offerings. Specifically, the overall SLA violation in a cloud data center at time t , denoted by $SLAV_t$, can be calculated by the following equation as proposed in [15].

$$SLAV_t = SLATAH_t \times PDM_t, \quad (3)$$

where $SLATAH_t$, the SLAV Time per Active Host at time t , is defined as the percentage of time and can be formulated by Equation (4) to indicate how long active hosts have experienced the CPU utilization of 100%; PDM_t is defined as the Performance Degradation due to migrations at time t and can be formulated by Equation (5) respectively:

$$SLATAH = \frac{1}{N} \sum_{i=1}^N \frac{T_{s_i}}{T_{a_i}} \quad (4)$$

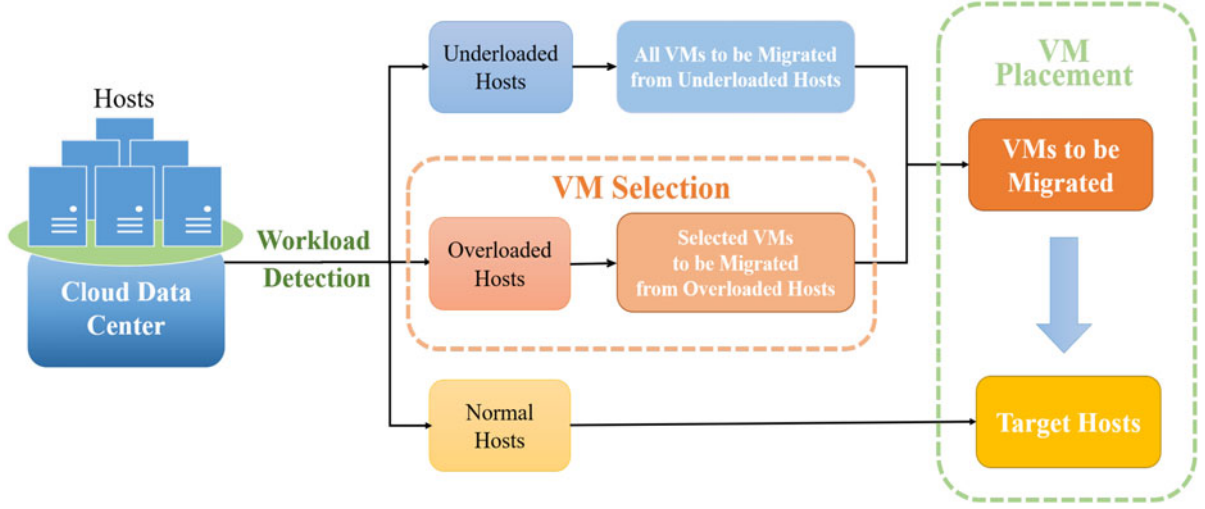


Fig. 1. Energy aware VM consolidation framework.

$$PDM = \frac{1}{n_i} \sum_{j=1}^{n_i} \frac{C_{d_j}}{C_{r_j}}, \quad (5)$$

in which T_{s_i} is the total time during which the host H_i has experienced the utilization of 100% leading to a SLAV, T_{a_i} is the total time of the host H_i being in the active state (serving VMs); n_i is the number of VMs on host H_i , C_{d_j} is the estimate of the performance degradation of the VM $V_{i,j}$ caused by migrations (in general, it is 10% of the CPU utilization in MIPS during all migrations of the VM $V_{i,j}$), and C_{r_j} is the total CPU capacity requested by the VM $V_{i,j}$ during its lifetime.

3.2 Energy Aware VM Consolidation Framework

The workloads of physical hosts often change dynamically during the operation of a cloud data center. If the utilization of certain resource, CPU or RAM for example, has reached maximum/minimum capacity of physical hosts, it will be at the risk of overload/underload at next moment, which will both lead to performance degradation and affect the quality of service for users. In this paper, we propose an Adaptive DRL-based VM Consolidation (ADVMC) framework to optimize resource utilization, which mainly focuses on reducing energy consumption while minimizing SLA violation.

The system framework of our ADVMC approach is illustrated in Fig. 1. First, hosts in a cloud data center are classified into three types by workload detection (usually static threshold-based or dynamic threshold-based methods according to previous researches): overloaded hosts, underloaded hosts and normal hosts whose workload are between overloaded hosts and underloaded hosts. Then VM selection and VM placement are included to realize energy aware VM consolidation:

- **VM Selection:** In this part, we focus on VM selection on overloaded hosts: a list of VMs will be selected for migration to restore the overloaded host to a normal state. In order to get rid of excessive workloads of overloaded host quickly and accurately, those VMs with the greatest impact on producing host overload will be excavated by considering Influence Coefficient. How to define Influence Coefficient and

how the Influence Coefficient-based VM selection works will be presented in Section 4.

- **VM Placement:** In this part, the VMs needed to be migrated, including all the VMs from underloaded hosts and selected VMs from overloaded hosts, will be placed onto suitable hosts. Therefore, a Prediction Aware DRL-based VM placement method (PADRL) is developed to find optimal policy automatically, in which a DRL-based VM placement method is proposed to map VMs and a state prediction network is specifically designed based on LSTM to speed up the convergence of DRL. How this LSTM-based prediction network works for DRL-based model and how the DRL-based VM placement strategy finally works will be discussed in Section 5.

4 INFLUENCE COEFFICIENT-BASED VM SELECTION

Since more than one VM run on a host in general, different VMs with different requirements will compete for the various resources on the same physical host. Therefore, each VM will have different impacts on producing host overload. In order to find the key factors causing host overload, so as to remove the excessive workloads of the overloaded host quickly and accurately, a novel Influence Coefficient is introduced in this part by calculating the relationship between the overloaded host and its VMs.

The Influence Coefficient, denoted by IC , is defined as follows:

$$IC(H_i, V_{i,j}) = \cos(H_i, V_{i,j}) \times (U_{V_{i,j}}^{CPU} U_{H_i}^{CPU} + U_{V_{i,j}}^{RAM} U_{H_i}^{RAM}), \quad (6)$$

where H_i is the overloaded host, and $V_{i,j}$ is the j th VM of H_i . $\cos(H_i, V_{i,j})$ is the cosine similarity between the resource usage of H_i and $V_{i,j}$, which can be obtained by equation (7):

$$\cos(H_i, V_{i,j}) = \frac{\sum_{r=1}^d U_{H_i}^r \times U_{V_{i,j}}^r}{\sqrt{\sum_{r=1}^d (U_{H_i}^r)^2} \times \sqrt{\sum_{r=1}^d (U_{V_{i,j}}^r)^2}}, \quad (7)$$

where d is the dimension of resources. On the basis of cosine similarity, we add the product of the resource usage

in equation (6) to further consider the volume of used resources. Here the resource usage indicates the amount of resources actually used. In our study, two typical resources, CPU and RAM, are included: $U_{H_i^{CPU}}$ and $U_{H_i^{RAM}}$ represent the CPU usage and RAM usage of the overloaded host H_i ; while $U_{V_{i,j}^{CPU}}$ and $U_{V_{i,j}^{RAM}}$ represent the CPU usage and RAM usage of the VM $V_{i,j}$, which are all normalized to a real number in the range 0 to 1 by the widely used min-max scaling. We believe that the greater the Influence Coefficient value is, the more similar between the resource usage of overloaded host and the VM is under all considered dimensions, which means more impact of the VM on the overloaded host. Therefore, to enlarge the different impact of VMs, we adopt the square of their resource usage in Equation (6).

Based on the definition of Influence Coefficient, a dynamic Influence Coefficient-based VM selection algorithm (ICVMS) is proposed to preferentially choose those VMs with the largest Influence Coefficient value, that is the VMs with the greatest impact, for migration. The process of our ICVMS method is presented in Algorithm 1.

Algorithm 1. Influence Coefficient-Based VM Selection

Input: an overloaded host H_i , thr // thr is predefined upper threshold for H_i
Output: a VM List for migration

```

1:  $migrateVMList \leftarrow NULL$ ;
2:  $hostUtil \leftarrow getResourceUtilization(H_i)$ ;
3:  $VMList \leftarrow getVMList(H_i)$ ;
4: repeat
5:    $maxInfluence \leftarrow 0$ ;
6:   for each VM in  $VMList$  do
7:      $VMUtil \leftarrow getResourceUtilization(VM)$ ;
8:      $ICValue \leftarrow IC(H_i, VM)$ ;
      // using equation (6)
9:     if  $ICValue > maxInfluence$  then
10:       $maxInfluence \leftarrow ICValue$ ;
11:       $VMmax \leftarrow VM$ ;
12:       $VMUtilmax \leftarrow VMUtil$ ;
13:    $VMList.remove(VMmax)$ ;
14:    $migrateVMList.add(VMmax)$ ;
15:    $hostUtil \leftarrow hostUtil - VMUtilmax$ ;
16: until  $hostUtil \leq thr$ 
17: return  $migrateVMList$ 
  
```

In Algorithm 1, it takes an overloaded host H_i and its predefined upper threshold as inputs. Each iteration of the loop chooses one VM for migration until H_i is no more overloaded. Inside the *for* loop, $ICValue$ is obtained using Equation (6) to find the VM with maximal influence on the overloaded host H_i . Finally, a selected VM list $migrateVmList$ is returned for migration.

5 PREDICTION AWARE DRL-BASED VM PLACEMENT

The VMs needed to be migrated, including all the VMs from underloaded hosts and selected VMs from overloaded hosts, have been collected in VM selection phase. In this VM placement phase, we need to find the most suitable host and conduct the final VM-Host mapping for each VM in order to further cut down energy consumption and reduce SLA violation.

Considering that VM placement problem can be regarded as a Markov decision problem, deep reinforcement learning (DRL) method is applied to tackle this issue and to obtain optimal policy. However, slow convergence is an inevitable problem of DRL caused by extensive exploration. In addition, tentative actions may result in large deviations of the reward from the optimum [33]. In particular, they may lead to excessive use of undesirable host resources and degradation of service quality, which in turn brings about extra energy consumption. Therefore, a Prediction Aware DRL-based VM placement method (PADRL) is proposed in this paper for developing adaptive mechanism to achieve energy saving and SLA violation reduction in cloud data centers.

As shown in Fig. 2, our PADRL scheme is comprised of two main parts: LSTM-based State Prediction and DQN-based VM Placement. The LSTM-based state prediction model is designed to predict future system states according to the real-world historical data. On the basis of these predicted states, the DQN-based VM placement network will be well pre-trained and can be applied to the real environment of cloud data centers to develop a more effective runtime VM placement policy. Both of these two parts will be discussed in more detail in the forthcoming sections.

5.1 LSTM-Based State Prediction

For providing the DRL agent more reasonable environment states to accelerate the convergence, a state prediction model based on Long Short Term Memory (LSTM) is proposed to predict future system states and to perform offline pre-training as the first step. LSTM, a recurrent neural network (RNN) published in 1997 by Hochreiter *et al.* [34], addresses the vanishing and exploding gradient problems of conventional RNN and thus is often used in different time series forecasting applications. It is well-suited to learn from experience to classify, process and predict time series [35], [36]. In this part, LSTM network is introduced to capture the characteristics of the dynamic changes of resources over time in the cloud environment.

The LSTM-based state prediction model is given in Fig. 3. As shown in Fig. 3, two LSTM layers are first involved to make more accurate prediction for hosts, in which each LSTM is responsible for predicting the next state of each host based on its historical system states of l time periods/steps. After LSTM layers, a fully connected layer and another output layer are used to yield the final prediction. The output of the model is the predicted states of all the N hosts in current cloud data center for the next DQN-based VM placement model.

The real-world workload from Google Cluster Trace is used as the training data to build the LSTM-based state prediction model in our study. Since the workload of Google Cluster Trace is sampled every five minutes, we set the time step length for state prediction to be five minutes accordingly. Besides, we use Mean Squared Error (MSE) as the loss function to train the batches between the prediction and the ground truth, and we use back propagation to derive network gradients and update LSTM network parameters. After that, the LSTM-based prediction model can be applied to provide the DQN-based VM placement model more reasonable environment states for training: the next state is predicted

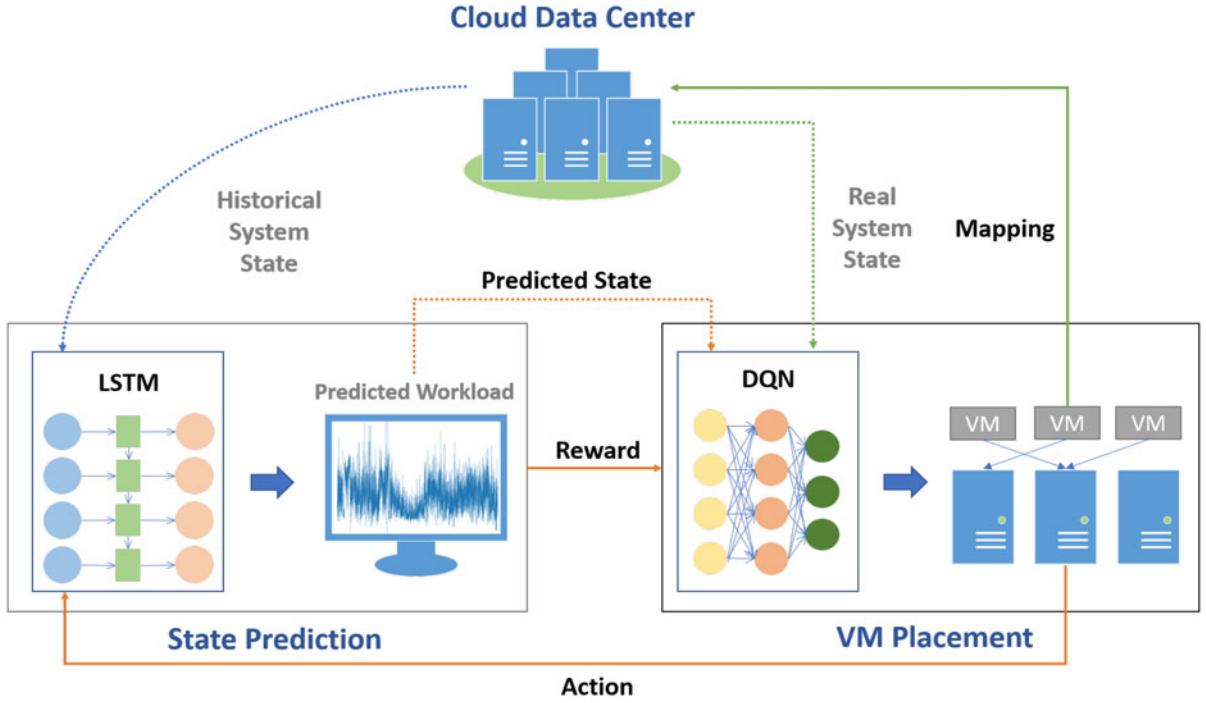


Fig. 2. Prediction aware DRL-based VM placement.

and fed to DQN-based model every time the selected VM placement action performing on the environment.

5.2 DQN-Based VM Placement

We now formulate the placement of VMs to be migrated as a deep reinforcement learning (DRL) problem, and a DQN-based VM placement method is used to find the optimal policy. The DQN-based VM placement model is given in Fig. 4. According to the current state of the cloud data center with heterogeneous hosts, an action of placing the VM on a suitable host will be selected and performed on the environment. Then a reward feedback for the impact of performing action will be received from the environment and the

environment, changed by the VM placement action, will move to a new state as a consequence.

5.2.1 Problem Model

The DQN-based method is used to develop an association between the best action (VM placement) and the state of the environment (cloud data center). Therefore, the state space, action space and the reward function are designed as follows:

The state space S represents the global status of the environment with each state representing the resource utilization of all hosts in a cloud data center. Therefore, a state at time t , denoted by s_t , can be defined as a tuple:

$$s_t = \langle u_1^t, u_2^t, \dots, u_k^t, \dots, u_N^t \rangle,$$

where u_k^t is the CPU utilization of host H_i . Here, only CPU performance is considered in order to make a comparative experiments with others. The limit of memory resource is

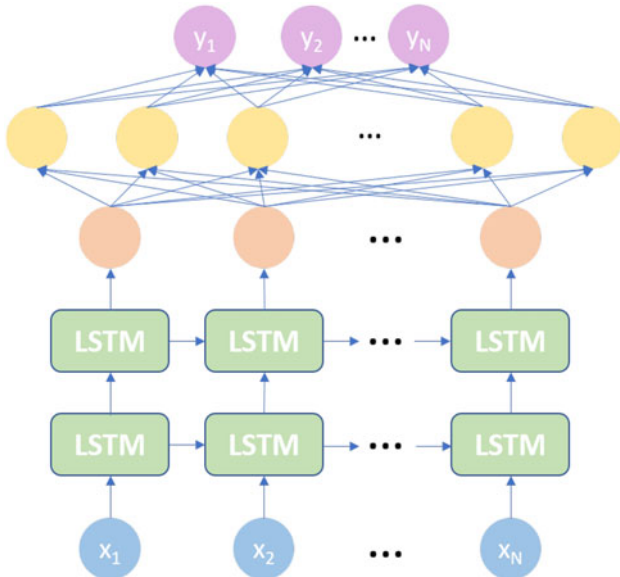


Fig. 3. LSTM-based state prediction model.

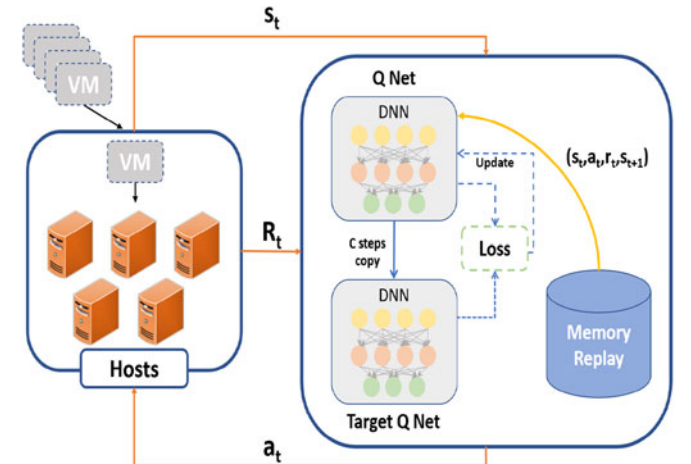


Fig. 4. DQN-based VM placement model.

used to determine whether the memory of target host meets the needs of VM at the final VM placement.

For finding a suitable physical host for each VM, the action space A is defined corresponding to the hosts in this paper. In this context, an action means to place the VM to a specific host in turn. Thus, an action at time t , denoted by a_t , is represented as a vector with each element indicating placed/not placed (1/0). Since each VM is considered individually, an action can have one and only one element taking the value of 1. For example, $a_t = (0, 0, 1, \dots, 0)$ refers to the action where the current VM under concern is to be placed to host H_3 at time t .

In this paper, reward is used to evaluate the efficiency of VM placement. With our goal to reduce energy consumption and minimize SLA violation, we construct the instant reward for each state updated by the sum of the energy consumption and the SLA violation before and after VM placement as follows:

$$r_t = (E_{t-1} - E_t) + (SLAV_{t-1} - SLAV_t), \quad (8)$$

where E_{t-1} and $SLAV_{t-1}$ are the energy consumption and the SLA violation at time $t-1$, and E_t and $SLAV_t$ are the energy consumption and the SLA violation at time t respectively. Here, energy consumption and SLAV at each time step will be processed first by scaling them to the same order of magnitude to eliminate the differences of data. Then the greater the value of the r_t is, the less energy consumption and SLA violation at the current moment is than previous time.

During the learning process, the DQN network is trained to capture the placement strategy by maximizing the expected return. It means that when making a decision through the Q-network, given the current state, only the action with the largest current Q value is required as the output action.

5.2.2 DQN-Based Algorithm

The process of our DQN-based VM placement method is presented in Algorithm 2.

In Algorithm 2, it takes parameters of Q-network θ , the exploration probability ϵ and the memory replay buffer $\mathcal{D}$ as inputs. $\mathcal{M}$ is the number of episode. The whole loop tries to find suitable host for each VM to be migrated. In each iteration of the loop, the agent first selects an placement action with exploration probability ϵ . After performing the action on the environment, the agent obtains a reward r_t from the environment on the basis of a predefined reward function, and then the environment moves to the next state. After that, the exploration probability ϵ will be decreased with the accumulated loops to reduce exploration and encourage exploitation step by step.

To improve the stability of convergence, target network $\hat{Q}$ is used as a supervisory signal to build loss with Q-network [37]. The parameters of target network, denoted by $\hat{\theta}$, will be initialized the same as the parameters of Q-network θ at the beginning. After performing a gradient descent step on loss function, it will be reset to θ every C time steps (C is a predefined number of steps to update target network). Furthermore, memory replay technique, proposed by Deepmind [33], is also introduced to alleviate the problems of correlated data and non-stationary distributions. In our DQN-based VM placement, the transition sequence (s_t, a_t, r_t, s_{t+1}) is stored

into a memory replay buffer $\mathcal{D}$ after interaction of agent and environment, and a random minibatch of transitions from memory replay buffer will be sampled for reducing data relevance.

Algorithm 2. DQN-Based VM Placement

Input: $\theta, \epsilon, \mathcal{D}, \mathcal{M}$
 // θ is parameters of Q-network
 // ϵ is the exploration probability
 // $\mathcal{D}$ is memory replay buffer
 // $\mathcal{M}$ is the number of episode
 1: Initialize Q-network with random weights θ
 2: Initialize target network $\hat{Q}$ with weights $\hat{\theta}: \hat{\theta} = \theta$
 3: Initialize $\epsilon, \mathcal{D}, \mathcal{M}$
 4: **for** $ep = 1$ to $\mathcal{M}$ **do**
 5: With probability ϵ select a random action a_t otherwise
 $a_t \leftarrow \operatorname{argmax}_a Q(s_t, a; \theta)$
 6: Obtain reward r_t and next state s_{t+1}
 7: Store transition (s_t, a_t, r_t, s_{t+1}) into memory replay buffer $\mathcal{D}$
 8: Sample random minibatch of transitions from memory replay buffer $\mathcal{D}$
 9: **if** *terminate at* s_{t+1} **then**
 10: $Q_{target} \leftarrow r_t$
 11: **else**
 12: $Q_{target} \leftarrow r_t + \gamma \max_{a_{t+1}} \hat{Q}(s_{t+1}, a_{t+1})$
 13: Perform a gradient descent step on $(Q_{target} - Q(s_t, a_t; \theta))^2$ with the weights θ
 14: Reset $\hat{\theta}$ to θ every C steps
 15: Update exploration probability ϵ

5.3 Algorithm Analysis

The complex cloud environment poses a significant challenge to the effective application of DRL methods in VM placement. Due to a large number of hosts and VMs involved in cloud data centers, the state-action space of DRL tends to be very huge. The DRL need to extensively explore the state-action space in the initial stage of training, and the aggressive action selection generated by the random exploration strategy may bring additional costs: the state of the environment transformed by the action of the next time step has a strong randomness since it is impossible to know the resource requirements of the VMs in the future. Thus, it is difficult to ensure the successful convergence of the decision-making model during the training process.

To address this problem, a LSTM-based state prediction model is proposed to pre-train the DQN-based VM placement network in this paper. LSTM-based state prediction model is designed to predict future states by the historical system states and thus can provide the DRL agent more reasonable environment states for training. Therefore, the agent can give the optimal VM placement action at run-time by forward-propagating the DRL based on the system state. As a result, the DRL-based VM placement model will enable an accelerated convergence by avoiding the instability of random initialization, and thus can learn a more successful VM placement policy with low run-time overhead.

6 EXPERIMENT AND EVALUATION

In this section, a series of experiments have been conducted on a real-world workload to evaluate the performance of

TABLE 1
Physical Machine Configuration

PM Type	Processor (MIPS)	Core	Memory (GB)
HP ProLiant G4	1860	2	4
HP ProLiant G5	2660	2	4

TABLE 2
Virtual Machine Configuration

VM Type	Processor (MIPS)	Memory (GB)
Micro instance	500	0.85
Small instance	1000	1.7
Extra-large Instance	2000	3.75
High-CPU Medium Instance	2500	0.85

our ADVMC approach, and to compare it with other VM selection and VM placement methods.

6.1 Experimental Setup

In order to verify the effectiveness of the proposed VM consolidation framework, we set up our experimental environment in CloudSim (Version 4.0) [38], a popular cloud simulation platform, to simulate a data center with 200 heterogeneous physical machines (hosts). The specific hardware configuration comes from the instance data in Amazon EC2 [39]: two different types of hosts and four types of VMs instances are included, with each host and each VM characterized by the CPU performance defined in millions instructions per second (MIPS) and amount of random access memory (RAM). The detailed configuration of the host and VM are shown in Tables 1 and 2 respectively.

As mentioned before, only CPU performance is considered in our energy consumption experiment, and a widely used SPECpower benchmark [31] is adapted in this paper. As shown in Table 3, the amount of power consumption of two types of hosts, HP ProLiant G4 (HP G4 in short) and HP ProLiant G5 (HP G5 in short), at different load levels are illustrated. Besides, we build a linear model to obtain a continuous power consumption for other CPU utilization between each ten percentage.

In this paper, a public and real-world workload from Google Cluster Trace is used for the operation of the host and VM. Google Cluster Trace provides real trace data of Google cluster collected every five minutes about one month period in May, 2011 [40]. We randomly extracts different day's workload data from Google Cluster Trace as our samples, and extracts a specified number of workload data from the chosen days for each experiment. The key characteristics of selected workload trace are presented in Table 4.

6.2 Performance Evaluation

In this section, we first compare the performance of our ICVMS method with different VM selection policies. Then we evaluate the performance of our PADRL method by focusing on the impact of LSTM-based state prediction model. Finally, we compare the overall performance of our ADVMC approach with other VM consolidation policies.

TABLE 3
Power Consumption at Different Load Levels in Watts

Host	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
HP G4	86	89.4	92.6	96	99.5	102	106	108	112	114	117
HP G5	93.7	97	101	105	110	116	121	125	129	133	135

TABLE 4
Characteristics of Selected Workload Trace

VMs	Resources	Mean (%)	Std.dev (%)	Median (%)
2877	CPU RAM	20.27 15.65	13.03 12.33	18 14

With our objectives on minimizing energy consumption and SLA violation, two metrics, energy consumption and SLAV introduced in 3.1, are used in this paper to evaluate the performance of different methods. All the results given below are the average of 10 experiments conducted independently.

Note that a static threshold-based method [38], for simplicity, is also adopted in our ADVMC approach to realize workload detection: two fixed thresholds are used to determine whether the host is overloaded or underloaded. In our experiments, the upper threshold for overloaded host is set to 0.9 and the under threshold for underloaded host is set to 0.2 for the best performance. Of course, some other state-of-the-art methods of workload detection could be applied here. However, it is not the concern of this paper and our previous studies and other related researches can be referred to.

6.2.1 Performance Evaluation of VM Selection

To evaluate the performance of the Influence Coefficient-based VM selection algorithm (ICVMS) for selecting VMs on overloaded hosts for migration, some widely used baselines proposed in [38] are employed in this part. 1) Maximum Utilization (MU): the VM with the maximum CPU utilization will be selected; 2) Minimum Migration Time (MMT): the VM that requires the minimum time to complete a migration will be selected; 3) Maximum Correlation (MC): the VM with the greatest correlation with other VMs on CPU utilization will be selected. The process of VM selection is executed according to different policies until the current host is not overloaded. Afterwards, the VM placement policy named PABFD proposed in [38] is also applied to obtain the final performance.

The average energy consumption and the average SLAV for various VM selection methods under different number of VMs are illustrated in Figs. 5a and 5a.

It can be seen that ICVMS policy can always produce the lowest energy consumption and SLAV no matter how the number of VMs changes. This benefits from the Influence Coefficient considered in ICVMS, which is able to select the VM with the greatest impact on producing overloaded host. In so doing, the overloaded host will possibly restore to the normal state faster by removing excessive workloads of overloaded host effectively, thereby reduces both the energy consumption and SLA violation. Moreover, the improvement of ICVMS becomes more remarkable compared to the

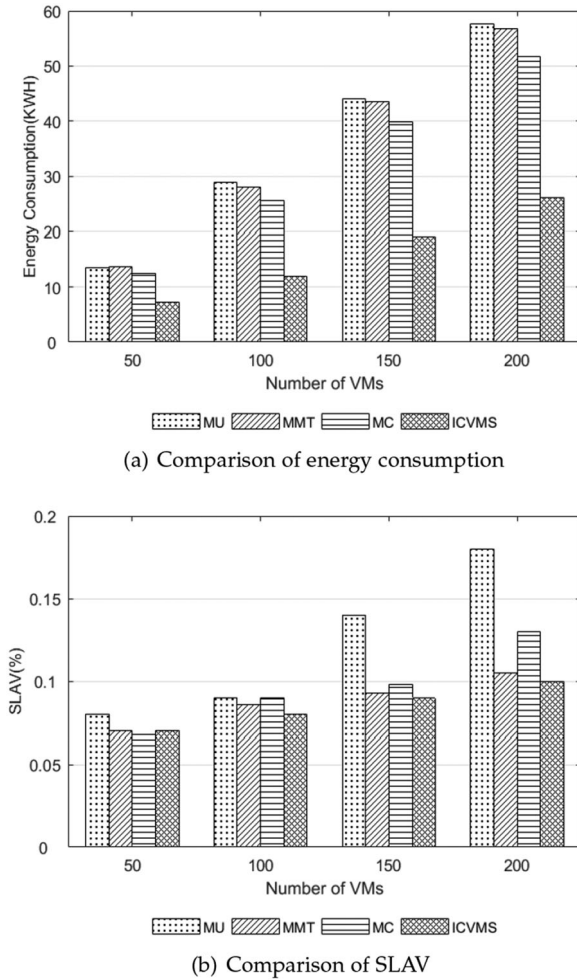


Fig. 5. Performance of ICVMS with different VM selection policies.

other three VM selection methods as the number of VMs increases. Overall, the average energy consumption and SLAV by ICVMS are reduced more than 112.30% and 38.33% respectively when the number of VMs increases to 200, which suggests that our ICVMS method is more suitable for the large scale environment of cloud data centers.

Furthermore, we conduct another experiment to compare ICVMS with two state-of-the-art methods: MRT and its improved algorithm MRT+MUP proposed in [21]. Here, the average improvement over three baselines (MMT, MC and MU) is compared due to the lack of source code. The performance comparison of ICVMS and these state-of-the-arts is given in Table 5.

As shown in Table 5, both two MRT methods and our ICVMS approach can achieve higher energy efficiency and SLA than three baselines. ICVMS remains producing a remarkable improvement on energy consumption by introducing Influence Coefficient in VM selection. It can also produce pretty good improvement on SLAV compared to MRT method. MRT+MUP performs slightly better on SLAV. It is not a surprise because MRT+MUP can estimate the future utilization of each resource type and thus can avoid unnecessary SLA violations with the help of multiple use of prediction scheme, which also provides an excellent point for our future research.

TABLE 5
Performance Comparison of ICVMS and two State-of-the-Arts

Methods	EC improvement over three baselines	SLAV improvement over three baselines
MRT	3.37%	0.33%
MRT+MUP	7.22%	40.18%
ICVMS	112.3%	38.33%

6.2.2 Performance Evaluation of VM Placement

In this set of experiments, we first focus on the performance of LSTM-based state prediction. The key parameter settings for LSTM-based state prediction model are given as below: The hidden size of the LSTM layers is set to 64. The sequence length l for each LSTM is set to 4. Four days' workload data chosen from Google Cluster Trace and another day's workload data are used as training set and testing set for LSTM-based state prediction model respectively. The proposed model will be trained for 100 epochs with batch size 128, using Adam optimizer and a learning rate of 0.01. Fig. 6 plots the convergence of LSTM-based state prediction model during the training process. It can be seen that the LSTM-based state prediction model will converge soon within 30 epochs by the real-world workload, and the Mean Square Error (MSE) of the model can reach 0.03 after training. Thus, it is able to predict more reasonable environment states for the DRL-based model.

Next, we further investigate the effectiveness of LSTM-based state prediction on the DRL-based VM placement. We compared two settings: with versus without LSTM-based state prediction model for PADRL. The former corresponds to our PADRL method based on prediction, whereas the latter, denoted by DRL, corresponds to the only DRL-based VM placement method without any pre-training. In this part, we construct Q-network by using a feed-forward neural network and a fully-connected neural network with two hidden layers (512 and 200 neurons for each hidden layer). Here we set the capacity of replay memory buffer to 5000 and minibatch size to 500. We update the Q-network parameters with the learning rate of 0.1. Parameters of target network are cloned from Q-network every 100 decision epochs. The discount factor γ is 0.9 and the exploration probability ϵ is decreased from 0.9 to 0 as the iteration increasing.

The experiment results about the average energy consumption and the average SLAV under different number of VMs are given in Figs. 7a and 7b.

It can be seen that PADRL always performs better than DRL as the number of VMs increased, on both the average energy consumption and the average SLAV. Overall, the average energy consumption of PADRL is reduced by 12.52%, and the average SLAV is reduced by 13.73% compared to DRL. This set of experiments has confirmed our analysis given in Section 5.3 that the DRL-based VM placement model can be well pre-trained by the predicted states from LSTM-based state prediction model, and will possibly achieve the optimal VM placement in a better and faster way. Therefore, this LSTM-based state prediction helps a lot to our PADRL method in cutting down system energy consumption and reducing SLAV.

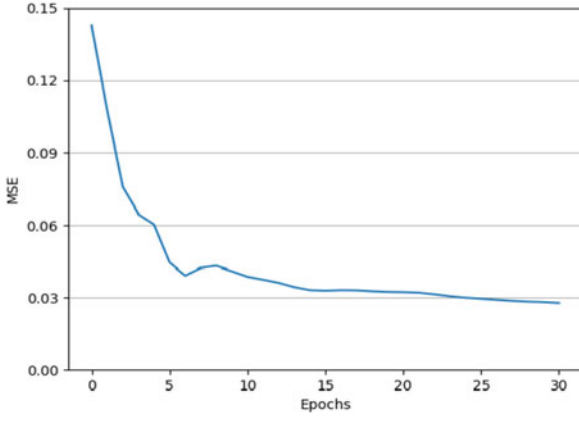
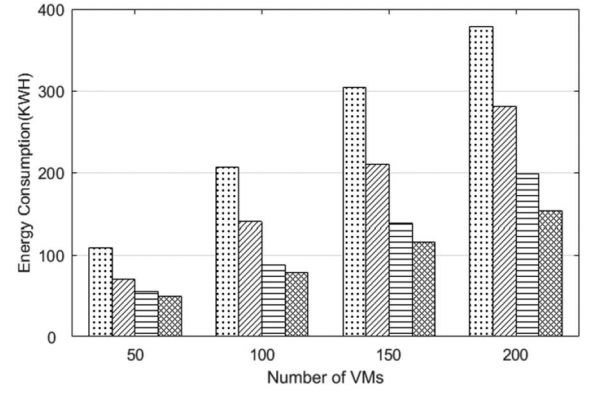


Fig. 6. The convergence of LSTM-based state prediction model during the training process.

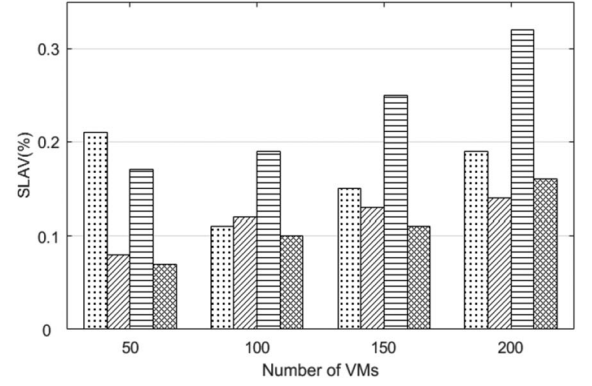
6.2.3 Performance Evaluation of VM Consolidation

In order to evaluate the overall performance of our Adaptive DRL-based Virtual Machine Consolidation (ADVMC) framework, we employ other three policies: 1) First Fit (FF), a typical widely studied heuristic method, proposed in [14], 2) Power Aware Best Fit Decreasing (PABFD), the default method in CloudSim, proposed in [38] and 3) another reinforcement learning (RL)-based method proposed in [41].

In this part, our VM selection method ICVMS is applied to all these policies to make a fair performance comparison, which means these policies, denoted by ICVMS-FF, ICVMS-PABFD and ICVMS-RL in short, will have the same VM selection settings as ICVMS. Fig. 8 plots the average energy

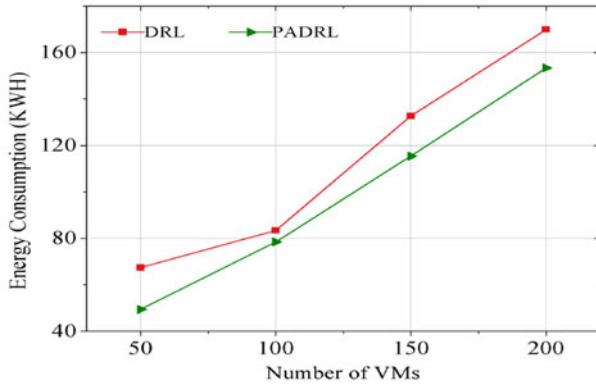


(a) Comparison of energy consumption

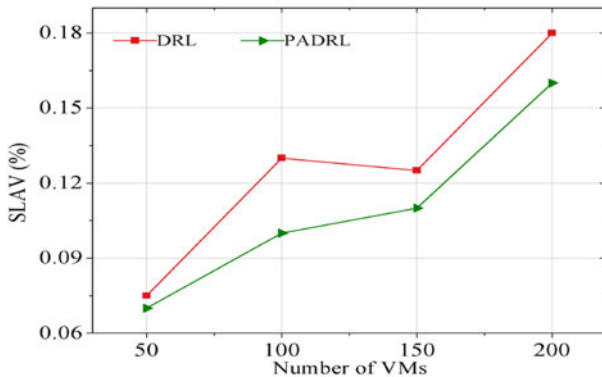


(b) Comparison of SLAV

Fig. 8. Performance of ADVMC with different VM consolidation policies.



(a) Comparison of energy consumption



(b) Comparison of SLAV

Fig. 7. Performance of PADRL under different number of VMs.

consumption and the average SLAV when the number of VMs increased from 50 to 200.

It can be seen that ICVMS-FF produces the highest energy consumption and pretty high SLAV since it always selects the first suitable host for VM placement without considering the current workload of the host. ICVMS-PABFD tries to select the host with the minimum increased energy consumption after VM placement, thus can produce better performance. ICVMS-RL can also produce good results by selecting the target host with the highest possible energy efficiency. However, the performance is not very stable especially when the space-action space becomes larger. In contrast, ADVMC consistently produces the least amount of energy consumption and SLAV among the policies considered, since ADVMC is designed with the objectives of cutting down energy consumption while minimizing the SLA violation.

Additionally, we also consider a latest research, named UP-VMC [20], in which a regression-based prediction model is utilized to approximate the future CPU and memory utilization. In order to make a comparative experiment, Modified First Fit Decreasing (MFFD) used in [20] is also implemented for comparison. The experiment results about the improvement over MFFD are plotted in Table 6.

It can be seen that both UP-VMC and ADVMC lead to significantly less energy consumption and SLA violation than MFFD. Overall, UP-VMC achieves 48.91% improvement of

TABLE 6
Performance Comparison of ADVMC and UP-VMC

Methods	EC improvement over MFFD	SLAV improvement over MFFD
UP-VMC	48.91%	123.71%
ADVMC	125.24%	138.64%

average energy consumption and 123.71% improvement of average SLAV over MFFD, while ADVMC achieves 125.24% improvement of average energy consumption and 138.64% improvement of average SLAV over MFFD. This confirms to us that our ADVMC approach performs better than UP-VMC especially for energy consumption. It benefits from the employment of two energy aware mechanisms in ADVMC: influence coefficient-based VM selection and prediction aware DRL-based VM placement, which are exactly the key factors of our ADVMC work contributed to minimize SLA violation and improve resource utilization for energy-efficient cloud data centers.

7 CONCLUSION AND FUTURE WORK

With the dramatic increasing of data and demands for computing capabilities, high energy consumption has become a growing concern in cloud data centers. Dynamic consolidation of VMs is proven to be an efficient way to tackle this issue. Therefore, it is critical to find viable solutions to VM consolidation so that resources of cloud data centers can be utilized in an energy-efficient way.

In this study, an adaptive VM consolidation mechanism based on DRL technique, named ADVMC, is investigated where VM selection and VM placement are both implemented to reduce energy consumption and to drop SLA violation. In particular, we first introduce a Influence Coefficient (IC) to measure the impact of a VM on producing host overload, and propose a dynamic Influence Coefficient-based VM selection algorithm (ICVMS) to preferentially choose those VMs with the greatest impact for migration. In so doing, excessive workloads of overloaded hosts can be removed more quickly and accurately. Next, we develop a Prediction Aware DRL-based VM placement method (PADRL) to find suitable hosts for VMs to be migrated with the promise of producing minimal energy consumption while providing a high level of adherence to SLAs. Based on a specific LSTM-based state prediction model, the DRL-based method can learn a more successful VM placement policy for energy-efficient cloud data centers in a faster way. Extensive experiments on real-world workload provided by Google Cluster Trace have confirmed that our ADVMC approach can largely cut down system energy consumption and reduce SLA violation as compared to other state-of-the-art models.

For future studies, we will attempt to optimize LSTM-based network structure by introducing more advanced network, such as Attention [42], to improve the prediction accuracy of environment states, and will investigate extended method to further improve the performance of DRL for energy-efficient VM consolidation in cloud data centers. Besides, we will take another key indicator, fault tolerance, into consideration to ensure the reliability of the cloud computing systems.

REFERENCES

- [1] M. Zhou, R. Zhang, D. Zeng, and W. Qian, "Services in the cloud computing era: A survey," in *Proc. 4th Int. Universal Commun. Symp.*, 2010, pp. 40–46.
- [2] P. Grover and R. Johari, "BCD: BigData, cloud computing and distributed computing," in *Proc. Glob. Conf. Commun. Technol.*, 2015, pp. 772–776.
- [3] S. Li, L. Da Xu, and S. Zhao, "5G internet of things: A survey," *J. Ind. Inf. Integration*, vol. 10, pp. 1–9, 2018.
- [4] S. Singh and I. Chana, "A survey on resource scheduling in cloud computing: Issues and challenges," *J. Grid Comput.*, vol. 14, no. 2, pp. 217–264, 2016.
- [5] M. Avgerinou, P. Bertoldi, and L. Castellazzi, "Trends in data centre energy consumption under the european code of conduct for data centre energy efficiency," *Energies*, vol. 10, no. 10, 2017, Art. no. 1470.
- [6] W. Deng, F. Liu, H. Jin, and D. Li, "Leveraging renewable energy in cloud computing datacenters: State of the art and future research," *Chin. J. Comput.*, vol. 36, no. 3, pp. 582–598, 2013.
- [7] G. Chen *et al.*, "Energy-aware server provisioning and load dispatching for connection-intensive internet services," in *Proc. 5th USENIX Symp. Netw. Syst. Des. Implementation*, 2008, pp. 337–350.
- [8] H. W. Tseng, T. Yang, K. Yang, and P. Chen, "An energy efficient VM management scheme with power-law characteristic in video streaming data centers," *IEEE Trans. Parallel Distrib. Syst.*, vol. 29, no. 2, pp. 297–311, Feb. 2018.
- [9] S. S. Masoumzadeh and H. Hlavacs, "An intelligent and adaptive threshold-based schema for energy and performance efficient dynamic VM consolidation," in *Proc. Eur. Conf. Energy Efficiency Large Scale Distrib. Syst.*, vol. 8046, 2013, pp. 85–97.
- [10] H. Liu, H. Jin, X. Liao, C. Yu, and C. Xu, "Live virtual machine migration via asynchronous replication and state synchronization," *IEEE Trans. Parallel Distrib. Syst.*, vol. 22, no. 12, pp. 1986–1999, Dec. 2011.
- [11] M. A. Kaaouache and S. Bouamama, "Solving bin packing problem with a hybrid genetic algorithm for VM placement in cloud," *Procedia Comput. Sci.*, vol. 60, pp. 1061–1069, 2015.
- [12] H. Zhao, J. Wang, F. Liu, Q. Wang, W. Zhang, and Q. Zheng, "Power-aware and performance-guaranteed virtual machine placement in the cloud," *IEEE Trans. Parallel Distrib. Syst.*, vol. 29, no. 6, pp. 1385–1400, Jun. 2018.
- [13] B. Speitkamp and M. Bichler, "A mathematical programming approach for server consolidation problems in virtualized data centers," *IEEE Trans. Services Comput.*, vol. 3, no. 4, pp. 266–278, Fourth Quarter 2010.
- [14] A. Beloglazov, J. Abawajy, and R. Buyya, "Energy-aware resource allocation heuristics for efficient management of data centers for cloud computing," *Future Gener. Comput. Syst.*, vol. 28, no. 5, pp. 755–768, 2012.
- [15] A. Beloglazov and R. Buyya, "Optimal online deterministic algorithms and adaptive heuristics for energy and performance efficient dynamic consolidation of virtual machines in cloud data centers," *Concurrency Comput.: Pract. Experience*, vol. 24, no. 13, pp. 1397–1420, 2012.
- [16] M. A. Khoshkholghi, M. N. Derahman, A. Abdullah, S. Subramaniam, and M. Othman, "Energy-efficient algorithms for dynamic virtual machine consolidation in cloud data centers," *IEEE Access*, vol. 5, pp. 10 709–10 722, 2017.
- [17] P. Guo, M. Liu, and Z. Xue, "A PSO-based energy-efficient fault-tolerant static scheduling algorithm for real-time tasks in clouds," in *Proc. IEEE 4th Int. Conf. Comput. Commun.*, 2018, pp. 2537–2541.
- [18] N. J. Kansal and I. Chana, "Energy-aware virtual machine migration for cloud computing—a firefly optimization approach," *J. Grid Comput.*, vol. 14, no. 2, pp. 327–345, 2016.
- [19] W. Zhang, X. Chen, and J. Jiang, "A multi-objective optimization method of initial virtual machine fault-tolerant placement for star topological data centers of cloud systems," *Tsinghua Sci. Technol.*, vol. 26, no. 1, pp. 95–111, 2021.
- [20] F. Farahnakian, T. Pahikkala, P. Liljeberg, J. Plosila, N. Hieu, and H. Tenhunen, "Energy-aware VM consolidation in cloud data centers using utilization prediction model," *IEEE Trans. Cloud Comput.*, vol. 7, no. 2, pp. 524–536, Second Quarter 2019.
- [21] N. T. Hieu, M. Di Francesco, and A. Ylä-Jääski, "Virtual machine consolidation with multiple usage prediction for energy-efficient cloud data centers," *IEEE Trans. Services Comput.*, vol. 13, no. 1, pp. 186–199, Jan./Feb. 2020.

- [22] S. Gupta and D. A. Dinesh, "Resource usage prediction of cloud workloads using deep bidirectional long short term memory networks," in *Proc. IEEE Int. Conf. Adv. Netw. Telecommun. Syst.*, 2017, pp. 1–6.
- [23] S. Thrun and M. L. Littman, "Reinforcement learning: an introduction," *AI Mag.*, vol. 21, no. 1, pp. 103–103, 2000.
- [24] N. C. Luong *et al.*, "Applications of deep reinforcement learning in communications and networking: A survey," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3133–3174, Fourth Quarter 2019.
- [25] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*. Cambridge, MA, USA: MIT Press, 2018.
- [26] D. Basu, X. Wang, Y. Hong, H. Chen, and S. Bressan, "Learn-as-you-go with megh: Efficient live migration of virtual machines," *IEEE Trans. Parallel Distrib. Syst.*, vol. 30, no. 8, pp. 1786–1801, Aug. 2019.
- [27] D. Ding, X. Fan, Y. Zhao, K. Kang, Q. Yin, and J. Zeng, "Q-learning based dynamic task scheduling for energy-efficient cloud computing," *Future Gener. Comput. Syst.*, vol. 108, pp. 361–371, 2020.
- [28] N. Liu, Z. Li, J. Xu, Z. Xu, S. Lin, Q. Qiu, J. Tang, and Y. Wang, "A hierarchical framework of cloud resource allocation and power management using deep reinforcement learning," in *Proc. IEEE 37th Int. Conf. Distrib. Comput. Syst.*, 2017, pp. 372–382.
- [29] Q. Zhang, M. Lin, L. T. Yang, Z. Chen, and P. Li, "Energy-efficient scheduling for real-time systems based on deep Q-learning model," *IEEE Trans. Sustain. Comput.*, vol. 4, no. 1, pp. 132–141, First Quarter 2019.
- [30] M. Cheng, J. Li, and S. Nazarian, "DRL-cloud: Deep reinforcement learning-based resource provisioning and task scheduling for cloud service providers," in *Proc. 23rd Asia South Pacific Des. Autom. Conf.*, 2018, pp. 129–134.
- [31] K. D. Lange, "Identifying shades of green: The specpower benchmarks," *Comput.*, vol. 42, no. 3, pp. 95–97, 2009.
- [32] W. Voorsluys, J. Broberg, S. Venugopal, and R. Buyya, "Cost of virtual machine live migration in clouds: A performance evaluation," *Cloud Comput.*, vol. 5931, pp. 254–265, 2009.
- [33] V. Mnih *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.
- [34] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [35] R. C. Staudemeyer and E. R. Morris, "Understanding LSTM—a tutorial into long short-term memory recurrent neural networks," 2019, *arXiv:1909.09586*.
- [36] D. Yi, X. Zhou, Y. Wen, and R. Tan, "Efficient compute-intensive job allocation in data centers via deep reinforcement learning," *IEEE Trans. Parallel Distrib. Syst.*, vol. 31, no. 6, pp. 1474–1485, Jun. 2020.
- [37] I. Osband, C. Blundell, A. Pritzel, and B. V. Roy, "Deep exploration via bootstrapped DQN," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst.*, vol. 29, 2016, Art. no. 4033–4041.
- [38] R. Calheiros, R. Ranjan, A. Beloglazov, C. De Rose, and R. Buyya, "Cloudsim: A toolkit for modeling and simulation of cloud computing environments and evaluation of resource provisioning algorithms," *Softw. Pract. Experience*, vol. 41, no. 1, pp. 23–50, 2011.
- [39] P. Barham *et al.*, "Xen and the art of virtualization," *ACM SIGOPS Operating Syst. Rev.*, vol. 37, no. 5, pp. 164–177, 2003.
- [40] C. Reiss, J. Wilkes, and J. L. Hellerstein, "Google cluster-usage traces: Format+ schema," *Google Inc., White Paper*, pp. 1–14, 2011.
- [41] F. Farahnakian, P. Liljeberg, and J. Plosila, "Energy-efficient virtual machines consolidation in cloud data centers using reinforcement learning," in *Proc. 22nd Euromicro Int. Conf. Parallel Distrib. Netw.-Based Process.*, 2014, pp. 500–507.
- [42] A. Vaswani *et al.*, "Attention is all you need," *Proc. Int. Conf. Neural Inf. Process. Syst.*, vol. 30, pp. 5998–6008, 2017. [Online]. Available: <https://proceedings.neurips.cc/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf>



Jing Zeng received the BEng degree in computer science and technology from Central China Normal University, in 2019. She is currently working toward the MS degree with the School of Computer and Information Technology, Beijing Jiaotong University. Her research interests include cloud computing, VM migration and workload prediction.



puting environment to enhance both the satisfaction of cloud users and the performance of whole cloud system.

Ding Ding received the PhD degree in computer application technology from Beijing Jiaotong University, in 2011. She is currently an associate professor with the School of Computer and Information Technology, Beijing Jiaotong University. With main research interests include cloud computing and parallel and distributed computing, she has published more than 30 related papers in recent years. Her current research interests include building highly efficient, fast and accurate resource management mechanism in cloud computing environment to enhance both the satisfaction of cloud users and the performance of whole cloud system.



Kaixuan Kang received the BS degree in software engineering from YanShan University, in 2018. She is currently working toward the PhD degree with the School of Computer and Information Technology, Beijing Jiaotong University. Her research interests include energy-efficient and accurate resource scheduling, deep reinforcement learning, effective representation learning for reinforcement learning and precise workload prediction.



HuaMao Xie received the BS degree in electronic information science and technology from Shangrao Normal University, in 2017, and the master's degree in computer technology engineering from the Jiangxi University of Science and Technology, in 2020. He is currently working toward the PhD degree with the Computer Science and Technology, Beijing Jiaotong University. His main research interests include wireless sensor network deployment, deep learning, cloud computing, and group intelligence optimization algorithms.



Qian Yin received the BS degree in information and computing science from Hebei GEO University, in 2019. She is currently working toward the MS degree with the School of Computer and Information Technology, Beijing Jiaotong University. Her research interests include energy-efficient resource scheduling in cloud computing and accurate workload prediction.

► For more information on this or any other computing topic, please visit our Digital Library at www.computer.org/csdl.